

Anatomy of Urinary Disease

Today's Materials

- Kidney Specimens
- Dissecting Equipment
- Kidney Blood Vessel Diagrams
- Scissors and Tape
- Models of urinary system organs
- Blank Pages in Lab Notebook
- Extra Textbooks/Manuals in Room

Activity 1: Kidney Anatomy & Diseases

In this activity, you will dissect a kidney specimen and identify regions of the kidney corresponding to a variety of urinary system diseases and conditions. You will specifically become familiar with the following anatomical parts:

Key Kidney Parts/Regions

- Fibrous capsule
 - Ureter
 - Renal vein/artery
 - Cortex
 - Medulla
 - Pyramids
 - Columns
 - Pelvis
 - Calices
1. Go to the diagram found on a later page of these activity sheets and label the 9 key kidney parts/regions listed above using some of the provided leader lines (you will not need to use all of the lines). Use the books in the room and web resources to assist you.
 2. Obtain a kidney specimen for your table, along with dissecting tools and safety equipment.
 3. Observe the kidney from the outside and identify any of the key kidney parts visible from this perspective.
 4. Now carefully cut the kidney in half along a frontal/coronal plane. That is, make your kidney look like the kidney diagram in these activity sheets.
 5. Identify the remainder of the key kidney parts that are visible in this view.

Kidney Diseases/Conditions

1. You will now have a chance to get to know seven different kidney diseases or conditions and relate those conditions to kidney anatomy. For each one you will

do some web research to understand the disease and then will relate the disease to kidney anatomy.

2. Perform web research on each of the following diseases/conditions and complete the tasks described after each one. Wikipedia is a great resource for information on these diseases/conditions, though you might need to branch out from Wikipedia to answer all the questions about certain diseases/conditions.
3. Note that your instructor will eventually assign your group one of these diseases/conditions to present in front of the class (see below following disease instructions)!

Disease/Condition 1: Renal Stenosis

As you conduct your research on this disease/condition, complete the following tasks and record answers/information in your notebook as appropriate.

1. In brief, what causes this disease/condition?
2. What symptoms or complications occur due to this disease/condition?
3. Which of the 9 key kidney parts/regions are most directly involved in this disease/condition?
4. On the diagram below, write the name of this disease (or an abbreviation for it) on top of the regions most involved in the disease.
5. How would those parts/regions look different in this disease/condition?
6. Find those same parts on your kidney specimen. Discuss how they currently look in your specimen compared to how they would be different in this disease.

Disease/Condition 2: MCKD (Medullary Cystic Kidney Disease)

As you conduct your research on this disease/condition, complete the following tasks and record answers/information in your notebook as appropriate.

1. In brief, what causes this disease/condition?
2. What symptoms or complications occur due to this disease/condition?
3. Which of the 9 key kidney parts/regions are most directly involved in this disease/condition?
4. On the diagram below, write the name of this disease (or an abbreviation for it) on top of the regions most involved in the disease.
5. How would those parts/regions look different in this disease/condition?
6. Find those same parts on your kidney specimen. Discuss how they currently look in your specimen compared to how they would be different in this disease.

Disease/Condition 3: Cortical Necrosis

As you conduct your research on this disease/condition, complete the following tasks and record answers/information in your notebook as appropriate.

1. In brief, what causes this disease/condition?
2. What symptoms or complications occur due to this disease/condition?
3. Which of the 9 key kidney parts/regions are most directly involved in this disease/condition?
4. On the diagram below, write the name of this disease (or an abbreviation for it) on top of the regions most involved in the disease.
5. How would those parts/regions look different in this disease/condition?

6. Find those same parts on your kidney specimen. Discuss how they currently look in your specimen compared to how they would be different in this disease.

Disease/Condition 4: Kidney Stones

As you conduct your research on this disease/condition, complete the following tasks and record answers/information in your notebook as appropriate.

1. In brief, what causes this disease/condition?
2. What symptoms or complications occur due to this disease/condition?
3. Which of the 9 key kidney parts/regions are most directly involved in this disease/condition?
4. On the diagram below, write the name of this disease (or an abbreviation for it) on top of the regions most involved in the disease.
5. How would those parts/regions look different in this disease/condition?
6. Find those same parts on your kidney specimen. Discuss how they currently look in your specimen compared to how they would be different in this disease.

Disease/Condition 5: Urinoma

As you conduct your research on this disease/condition, complete the following tasks and record answers/information in your notebook as appropriate.

1. In brief, what causes this disease/condition?
2. What symptoms or complications occur due to this disease/condition?
3. Which of the 9 key kidney parts/regions are most directly involved in this disease/condition?
4. On the diagram below, write the name of this disease (or an abbreviation for it) on top of the regions most involved in the disease.
5. How would those parts/regions look different in this disease/condition?
6. Find those same parts on your kidney specimen. Discuss how they currently look in your specimen compared to how they would be different in this disease.

Disease/Condition 6: Renal Pseudotumors

As you conduct your research on this disease/condition, complete the following tasks and record answers/information in your notebook as appropriate.

1. In brief, what causes this disease/condition?
2. What symptoms or complications occur due to this disease/condition?
3. Which of the 9 key kidney parts/regions are most directly involved in this disease/condition?
4. On the diagram below, write the name of this disease (or an abbreviation for it) on top of the regions most involved in the disease.
5. How would those parts/regions look different in this disease/condition?
6. Find those same parts on your kidney specimen. Discuss how they currently look in your specimen compared to how they would be different in this disease.

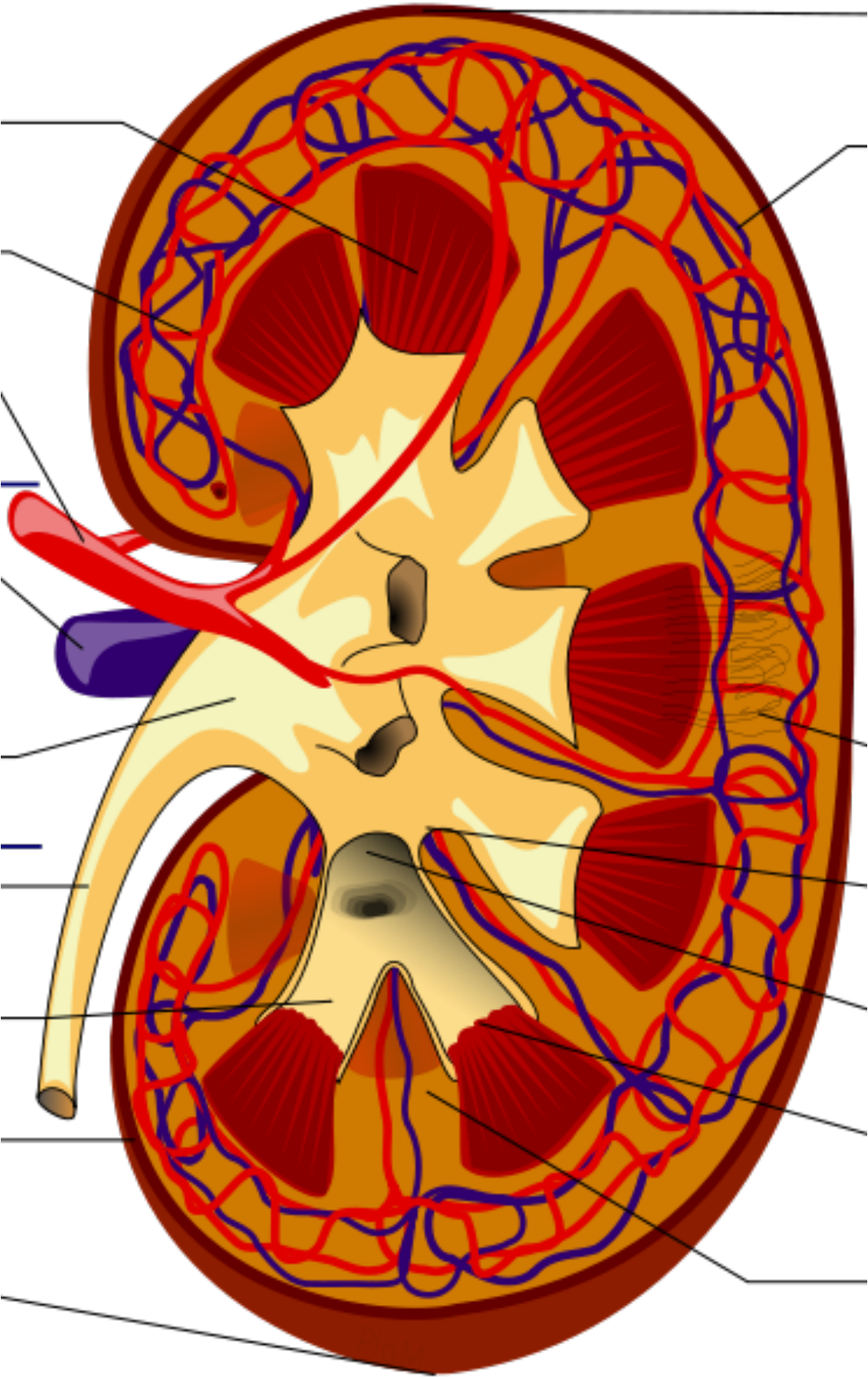
Disease/Condition 7: Increased Echogenicity of Renal Pyramids

As you conduct your research on this disease/condition, complete the following tasks and record answers/information in your notebook as appropriate.

1. In brief, what causes this disease/condition?
2. What symptoms or complications occur due to this disease/condition?
3. Which of the 9 key kidney parts/regions are most directly involved in this disease/condition?
4. On the diagram below, write the name of this disease (or an abbreviation for it) on top of the regions most involved in the disease.
5. How would those parts/regions look different in this disease/condition?
6. Find those same parts on your kidney specimen. Discuss how they currently look in your specimen compared to how they would be different in this disease.

Disease Presentations

1. You should have now learned about all 7 diseases/conditions above, but your instructor will now assign you one disease to present to the class.
2. You will bring your kidney specimen to the front of the room and display it using the document projector. As you discuss your assigned disease/condition, you will point to and name the relevant parts on your specimen.
3. Your *brief* presentation should include:
 - a. The name of the disease/condition
 - b. The kidney part(s) most directly involved (while pointing to them), and how those parts behave/look differently in the disease/condition.
 - c. The symptoms or complications associated with the disease/condition.
4. When not presenting yourself, you should listen carefully to the other presenters, supplement your notes regarding the disease/condition being presented, and ask questions to clarify information, particularly if you found different or conflicting information when you researched the disease/condition.



Activity 2: Tracing Kidney Blood Flow

The kidneys are rich in blood vessels and receive approximately 25% of all the heart's output of blood! This is perhaps not surprising, given the kidneys' roles in filtering the blood, balancing the body's fluids and solutes, and maintaining appropriate blood pressure. The extraordinarily dense blood vessels in the kidney are intricately arranged to extend from the aorta to the functional physiological unit of kidneys – the nephrons – and back to the vena cava. In this activity, we will trace the path of blood across these vessels and explore the significance of that pathway.

1. Use scissors to cut out the blood vessel names below.
2. Obtain one of the laminated printouts of blood vessels in the kidney.
3. Use information from internet resources and/or books to tape the blood vessel names on to the corresponding vessels on the laminated printout.
4. Check with your instructor to determine whether the labels are correct. Once correct, take a picture (or record notes or a diagram) to help you study the names and locations later.

Segmental Artery	Arcuate Vein
Cortical Radiate Artery	Renal Vein
Peritubular Capillaries	Efferent Arteriole
Interlobar Vein	Renal Artery
Glomerulus Capillaries	Afferent Arteriole
Cortical Radiate Vein	Interlobar Artery
Arcuate Artery	Vasa Recta

5. Sometimes blood flow in vessels can be blocked due to atherosclerosis, clots, or other conditions. Blockages will increase blood pressure (BP) on the afferent side of the obstruction and decrease BP on the efferent side.
6. The boxes below represent different blood vessel blockages. Cut out the boxes and place them on the diagram to indicate where a blockage would create the problems described.

Blockages:

<i>Increases BP Going Into Kidney & In Glomerulus</i>
<i>Decreases BP Going Into Kidney & In Glomerulus</i>
<i>Increases BP Going Into Kidney, Decreases BP In Glomerulus</i>
<i>Increases BP in Efferent Arteriole, Decreases BP in Vasa Recta</i>
<i>Increases BP in Interlobar Artery, Decreases BP in Cortical Radiate Arteries</i>

7. Check with your instructor to determine whether the blockage locations are correct. Once correct, take a picture (or record notes or a diagram) to help you remember your work.

Activity 3: The Bladder & Urethra

1. Identify the following structures on bladder models in the room:
 - a. Ureter
 - b. Detrusor muscle
 - c. Uretral openings
 - d. Internal urethral sphincter
 - e. External urethral sphincter
 - f. Urethra
2. Perform some internet research on differences in urethra anatomy between males and females. Record down the differences you uncover. How could you tell the difference between a male and female anatomical diagram of the bladder and urethra?
3. Based on that research (and additional research you conduct), explain...
 - a. The anatomical reasons that explain why females are much more prone than males to getting urinary tract infections.
 - b. The anatomical reasons that explain why older males (more so than females) often have slow urination, frequent urination, or difficulty urinating.
 - c. The anatomical reasons that explain why urine abnormalities might be more likely to impact fertility in males than in females.